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INFORMATION SECURITY LAB

MINI PROJECT – SYNOPSIS

1. SecurePDF: A Cryptographic PDF Fingerprinting and Intelligent Document Forensics System

2. Team Details

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3. Introduction / Functionality

SecurePDF is a secure document distribution and forensic analysis system designed to identify the intended recipient of leaked PDF documents and assist in analyzing unauthorized modifications. Traditional visible watermarks can be easily removed or altered and do not provide reliable cryptographic verification of a document's recipient.

The proposed system generates a unique, recipient-specific fingerprint for each distributed PDF using HMAC-SHA256 and embeds it in two locations within the PDF structure: the document catalog and invisible annotations. A SQLite database maintains the corresponding fingerprint records, enabling automated verification and recipient attribution when a suspicious document is submitted for analysis.

The system also incorporates a Machine Learning (ML)-based forensic analysis module that extracts structural features from PDFs and classifies them into categories such as original, benignly transformed, modified, and marker-tampered documents. A user-friendly command-line interface supports document issuance, verification, and forensic analysis, while a demonstration interface facilitates batch processing, controlled tampering experiments, visual comparisons, and automated report generation.

SecurePDF combines cryptographic authentication, redundant fingerprint storage, automated document analysis, and visual forensic reporting into a unified framework for document-leak investigation.

4. Objectives

1. PDF Fingerprint Generation and Issuance

Generates a unique fingerprint using document identifiers, recipient identifiers, a nonce, and HMAC-SHA256 with a secret master key. Each recipient receives a personalized PDF.

2. Dual-Layer Fingerprint Embedding

Embeds fingerprint payloads in two locations: the PDF catalog (Layer 1) and invisible PDF annotations (Layer 2). The redundant design improves resilience when one carrier is damaged or removed.

3. Cryptographic Verification and Recipient Attribution

Extracts fingerprint payloads, retrieves candidate records from SQLite, recomputes the HMAC, and verifies authenticity before attributing a document to a recipient. It distinguishes valid fingerprints, invalid fingerprints, missing fingerprints, and conflicting authenticated identities.

4. PDF Structural Feature Extraction

Uses PyMuPDF to extract structural features, including page count, metadata, annotations, embedded files, JavaScript/action indicators, and fingerprint-carrier characteristics.

5. Machine Learning-Based Forensic Classification

Uses a Random Forest classifier to analyze controlled PDF examples categorized as original, benignly transformed, modified, or marker-tampered. Model performance is evaluated using accuracy, precision, recall, F1-score, confusion matrices, and a dummy-classifier baseline.

6. Controlled Tampering and Transformation Testing

Generates experimental copies of issued PDFs with transformations such as metadata changes, page addition or deletion, reordering, content modification, and fingerprint-carrier corruption or removal. The system compares verification outcomes before and after each transformation.

7. Visual Forensic Analysis and Reporting

Uses PyMuPDF and Matplotlib/Seaborn to generate page comparisons, difference visualizations, confusion matrices, and evaluation charts where applicable. Automated reports summarize cryptographic outcomes, detected structural differences, and ML findings.

8. Database and Key Management

Uses SQLite to maintain document, recipient, and fingerprint records. A secret master key protects fingerprint authentication, while isolated experimental environments support safe testing.

9. User Interface and Demonstration CLI

Provides an interactive CLI for managing students, selecting input PDFs, issuing personalized copies, analyzing suspicious documents, running ML evaluations, and generating reports and charts.

Information Security techniques implemented: HMAC-SHA256-based message authentication, cryptographic fingerprinting, nonce-based uniqueness, redundant fingerprint embedding, integrity-oriented verification, and forensic analysis of document modifications.

5. Proposed Functionality / Modules

Briefly describe the major modules/functions of the project. Clearly mention the Information Security technique/algorithm used and implemented.

6. Technology / Tools Used

Programming Language	Python
Software / Tools	VS Code, Gitm PyMuPDF
Database (if applicable)	SQLite
Security Algorithm / Technique	HMAC-SHA256, secret-key authentication, nonce-based fingerprinting, dual-layer PDF fingerprint embedding
Other Technologies	Scikit-learn, Random Forest, Dummy Classifier, numpy, Pandas, seaborn, matplotlib

7. Expected Outcome

The completed SecurePDF system provides an automated approach to personalized PDF distribution, cryptographic recipient attribution, and document-forensics experimentation.

The system is designed to:

- Generate uniquely fingerprinted PDF copies for multiple recipients.
- Authenticate extracted fingerprints using HMAC-SHA256 and identify the corresponding recipient when verification succeeds.
- Detect invalid fingerprints, missing carriers, and conflicting authenticated identities through explicit verification outcomes.
- Analyze controlled PDF transformations using structural features and an experimental Machine Learning classifier.
- Produce visual comparisons, evaluation charts, and structured forensic reports.
- Demonstrate the complementary roles of cryptographic authentication and ML-based analysis in investigating document leaks.

Important limitation: A valid HMAC establishes fingerprint authenticity and supports recipient attribution; it does not, by itself, prove that the PDF's content is unchanged. Similarly, if both fingerprint carriers are removed, the system cannot establish prior issuance from the PDF alone without an independent trusted issuance record. ML findings are advisory and do not override cryptographic verification.

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Date